GATE ME 90 Days Challenge

Day 21/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
General Aptitude	**Quantitative Aptitude I**	**87 days to GATE**

How to use this set: 15 questions (5 MCQ, 10 NAT) on percentages, ratio and proportion, averages, profit and loss, time-speed-distance (trains, boats, races) and time and work (pipes, efficiency, man-days). Try them in **30 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A's monthly income is 25 % less than B's. By what percentage is B's income more than A's?

(A) 25 %
(B) 20 %
(C) 33.33 %
(D) 30 %

Q2 NAT

The price of an item is first increased by 10 % and then decreased by 20 %. The net percentage change in its price is (enter a negative number for a decrease) _____

Q3 MCQ

₹7800 is divided among A, B and C in the ratio 1/2 : 1/3 : 1/4. A's share is

(A) ₹3900
(B) ₹3600
(C) ₹2600
(D) ₹3000

Q4 NAT

The average of 11 results is 60. The average of the first six results is 58 and that of the last six is 63. The sixth result is _____

Q5 NAT

A batsman's average after 10 innings is 32. How many runs must he score in the 11th innings to raise his average to 36? _____

Q6 MCQ

An article costs ₹800. The shopkeeper wants a 15 % profit after allowing an 8 % discount on the marked price. The marked price must be

(A) ₹1000
(B) ₹920
(C) ₹1012
(D) ₹994

Q7 NAT

A dishonest dealer claims to sell at cost price but uses a weight of 800 g instead of 1 kg. His gain percentage is _____

Q8 NAT

A train passes a pole in 15 seconds and a 100 m long platform in 25 seconds. The length of the train (in metres) is _____

Q9 NAT

Two cars start at the same time from towns P and Q, 300 km apart, and drive towards each other at 40 km/h and 60 km/h respectively. The distance of their meeting point from P (in km) is _____

Q10 MCQ

A boat's speed in still water is 8 km/h and the stream flows at 2 km/h. The time taken to go 30 km downstream and come back is

(A) 7.5 h
(B) 8 h
(C) 6 h
(D) 10 h

Q11 NAT

12 men can complete a work in 18 days. They work for 6 days, and then 4 more men join them. The number of days needed to finish the remaining work is _____

Q12 NAT

A is twice as efficient as B. Working together they finish a job in 14 days. The number of days A alone would take is _____

Q13 MCQ

A pipe can fill a tank in 6 hours. Because of a leak at the bottom, it takes 8 hours to fill the tank. The leak alone can empty the full tank in

(A) 14 h
(B) 2 h
(C) 48 h
(D) 24 h

Q14 NAT

A 40-litre mixture contains milk and water in the ratio 3 : 1. How much water (in litres) must be added to make the ratio 3 : 2? _____

Q15 NAT

In a 200 m race, A beats B by 20 m and A beats C by 38 m. In a 200 m race between B and C, B beats C by (in metres) _____

Answer Key

Q	Type	Answer
Q1	MCQ	(C) 33.33 %
Q2	NAT	-12
Q3	MCQ	(B) ₹3600
Q4	NAT	66
Q5	NAT	76
Q6	MCQ	(A) ₹1000
Q7	NAT	25
Q8	NAT	150
Q9	NAT	120
Q10	MCQ	(B) 8 h
Q11	NAT	9
Q12	NAT	21
Q13	MCQ	(D) 24 h
Q14	NAT	10
Q15	NAT	20

Detailed Solutions

Q1 **Answer: (C) 33.33 %**

Concept: Percentage change is always measured on the base you compare against.

Let B = 100, then A = 75. B exceeds A by 25 on a base of 75.

25/75 × 100 = **33.33 %**

Exam tip: If A is r % less than B, then B is r/(100 − r) × 100 % more than A. Memorise the pairs 20 ↔ 25, 25 ↔ 33.33, 50 ↔ 100.

Q2 **Answer: −12**

Concept: Successive changes multiply: net factor = (1 + a/100)(1 + b/100).

1.10 × 0.80 = **0.88**

Net change = **-12 %** (a 12 % decrease)

Exam tip: Shortcut a + b + ab/100 = 10 − 20 − 2 = −12. The order of the changes does not matter.

Q3 **Answer: (B) ₹3600**

Concept: Clear fractional ratios by multiplying every term by the LCM of the denominators.

LCM(2, 3, 4) = 12 → ratio = 6 : 4 : 3 (sum 13)

A = 6/13 × 7800 = **₹3600**

Exam tip: Check: B = ₹2400 and C = ₹1800; 3600 + 2400 + 1800 = 7800.

Q4 **Answer: 66**

Concept: The sixth result is counted in both groups, so adding the two group-sums counts it twice.

Sum of all 11 = 660; first six = 348; last six = 378

Sixth = 348 + 378 − 660 = **66**

Exam tip: Overlap problems: (sum of groups) − (total) = the overlapped item(s).

Q5 **Answer: 76**

Concept: Work with totals, not averages.

Total after 10 innings = 320; required total after 11 = 11 × 36 = 396

Runs needed = 396 − 320 = **76**

Exam tip: Shortcut: new score = new average + (number of old innings × rise) = 36 + 10 × 4 = 76.

Q6 **Answer: (A) ₹1000**

Concept: SP = CP(1 + profit %) = MP(1 − discount %).

SP = 800 × 1.15 = **₹920**

MP = 920/0.92 = **₹1000**

Exam tip: MP/CP = (1 + p)/(1 − d) = 1.15/0.92 = 1.25: mark the article 25 % above cost.

Q7 **Answer: 25**

Concept: Gain % = (true weight − false weight)/false weight × 100.

He gives goods costing ₹800 but charges ₹1000.

Gain = 200/800 × 100 = **25 %**

Exam tip: Divide by the false weight (800), not the true weight (1000). That trap gives the common wrong answer 20 %.

Q8 **Answer: 150**

Concept: Passing a pole: distance = train length; passing a platform: distance = train length + platform length.

The extra 10 s is spent covering the 100 m platform, so speed = 100/10 = **10 m/s**

Length = 10 × 15 = **150 m**

Exam tip: Speed = 10 m/s = 36 km/h. Multiply m/s by 18/5 to convert to km/h.

Q9 **Answer: 120**

Concept: Moving towards each other, the gap closes at the sum of the speeds.

Time to meet = 300/(40 + 60) = **3 h**

Distance from P = 40 × 3 = **120 km**

Exam tip: The meeting point divides the distance in the ratio of the speeds, 40 : 60 = 2 : 3, so 2/5 × 300 = 120 km without finding the time.

Q10 **Answer: (B) 8 h**

Concept: Downstream speed = b + s, upstream speed = b − s.

Downstream: 30/10 = 3 h; upstream: 30/6 = 5 h

Total = **8 h**

Exam tip: 30 × 2/8 = 7.5 h is the answer in still water: a current always lengthens a round trip.

Q11 **Answer: 9**

Concept: Use man-days: work = men × days.

Total work = 12 × 18 = 216 man-days; done in 6 days = 72 man-days

Remaining 144 man-days with 16 men: 144/16 = **9 days**

Exam tip: Total time = 6 + 9 = 15 days. Man-day accounting avoids fractions completely.

Q12 **Answer: 21**

Concept: Efficiency is the work done per day; time is inversely proportional to efficiency.

Let B's rate = x, A's rate = 2x. 3x × 14 = 1 → x = 1/42

B alone: 42 days; A alone (twice as fast): **21 days**

Exam tip: Ratio of rates 2 : 1, so A does 2/3 of the work in 14 days; the whole work takes A 14 × 3/2 = 21 days.

Q13 **Answer: (D) 24 h**

Concept: Net rate = filling rate − leak rate.

Leak rate = 1/6 − 1/8 = (4 − 3)/24 = 1/24 of the tank per hour

Leak empties the tank in **24 h**

Exam tip: Shortcut: ab/(b − a) = 6 × 8/(8 − 6) = 24 h.

Q14 **Answer: 10**

Concept: The quantity that does not change (milk) fixes the new total.

Milk = 30 L, water = 10 L.

For 3 : 2, water = 30 × 2/3 = 20 L, so add 20 − 10 = **10 L**

Exam tip: Always anchor on the unchanged component; never scale the whole mixture.

Q15 **Answer: 20**

Concept: In the same time the distances run are in the ratio of the speeds.

When A runs 200 m: B runs 180 m and C runs 162 m. So B : C = 180 : 162 = 10 : 9.

When B runs 200 m, C runs 200 × 9/10 = 180 m: B wins by **20 m**

Exam tip: Not 38 − 20 = 18 m: the margins come from different race lengths, so use speed ratios.

What next?

Day 21 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 22/90: Vibrations, Forced Vibration. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.